D (Printed Pages 4)

(21223) Roll No. .....

BCA-III Sem.

18012

B.C.A. Examination, Dec.-2023

Data Structure using C and C++

(BCA-302)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** the Sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt all **five** questions. Each question carry 3 marks. 3×5=15

1. What do you mean by Sparse arrays? Give example. 3
2. Where can we use Stack? Give applications of stack. 3
3. How can you represent a Binary Tree in memory using array? 3

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4. What is Hashing. What are its advantages. 3
5. Explain Priority Queue in brief. 3

Section-B

(Short Answer Question)

Note : Attempt any **two** questions.

6. Write an algorithm to convert a given infix expression to postfix expression. Trace the steps involved in converting the given infix expression ((A+B)^C)-((D*C)/F) to post fix expression. 7.5
7. Give difference between Linear and Binary Search. Also write a C program to implement Binary Search. 7.5
8. Write C program to show insertion and deletion in a simple Queue. 7.5

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Section-C

(Long Answer Question)

Note : Attempt any **three** questions.

3×15=45

9. Perform Heap Sort on given keys, Frrange them in descending order: 15

K=2, 9, 7, 6, 5, 8

10. Construct a Binary Tree from its 15

Inorder Traversal → D, B, E, A, F, C

Preorder Traversal → A, B, D, E, C, F

11. Write a C Program to multiply two 2-Dimensional matrix of 3×3 and store the result in another matrix. 15

12. Implement Singly linked list using dynamic memory allocation. 15

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13. Write short notes on: 15

(a) Collision resolution techniques

(b) B-Trees

(c) D-Queues

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A (Printed Pages 3)

(21222) Roll No. _____

BCA-III Sem.

18012

B.C.A. Examination, Dec.-2022

Data Structure Using C and C++

(BCA-302)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt question from **all** sections as per instructions.

Section - A

(Very Short Answer Type Questions)

Note : Attempt all **five** questions. Each question carry 3 marks.

1. Explain how two dimensional array is stored in memory?
2. What is priority queue?
3. What is linked list?

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4. Explain B-tree.
5. What is hashing?

Section-B

(Short Answer Type Question)

Note : Attempt any two out of the following three. Each carry 7½ marks.

6. Explain stack by using linked list with suitable example.
7. Write a program of multidimensional array. (m×n)
8. Define structure and union with supportive example.

Section-C

(Long Answer Type Questions)

Note : Attempt **any** 3 questions out of 5 question. Each carry 15 marks.

9. (a) Explain the method of deletion in single linked list.

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(b) What is the sorting? Explain selection sort.

10. (a) What is hashing? Explain different type of hash function.

(b) What is Binary tree? How will you search in Binary tree.

11. What is tree. Explain type of trees if any. Give the traversal of Binary tree with algorith and supportive example of all traversal.

12. Explain heap sort and show step by step procedure to sort the data using heap sort technique.

13. (a) Differentiate Binary and linear search with supportive example.

(b) Discuss the properties of B-tree in detail.

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A (Printed Pages 4)

(20222) Roll No. .....

BCA-III Sem.

18012 (CV-III)

B.C.A. Examination, Dec.-2021

DATA STRUCTURE USING C AND C++

[BCA-302]

Time : 1½ Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** Sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt any **two** questions. Each question carry **7.5** marks.

2×7.5=15

1. What is hashing?
2. What is Binary Tree?
3. What is deque?
4. What is analysis of algorithm?

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5. Explain principle of Recursion?

Section-B

(Short Answer Type Questions)

Note : Attempt any **one** out of the following three questions. Each question carry **15** marks. 1×15=15

6. What is linked list? Explain with suitable example and give an algorithm for insert an element.
7. Explain heap sort with suitable example.
8. Define and explain the stack data structure with suitable example.

Section-C

(Long Answer Type Questions)

Note : Attempt any **two** questions out of the following five questions. Each question carry **22.5** marks.

2×22.5=45

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9. What is Binary Tree? Explain it. The following binary tree has the inorder and preorder traversal. Draw the tree and give the postorder traversal. Write algorithm for the same.

Inorder- ABCEDFJGIH

Preorder- JCBADEFIGH

10. Explain Merge sort sorting algorithm and show step by step procedure to sort the data.

11. Explain Binary search and linear search? Give the supportive example for both.

12. (a) What is Multi dimensional Array? Explain with example.

(b) Write the step by step for following expression.

R|D–x * (G/E * (A+B) + D–E * (Y–F)) + S*J

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13. (a) Why doubly linked list is better than linked list? Justify it with example.

(b) What is threaded Binary tree? Explain it with suitable example.

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D (Printed Pages 4)

(20221) Roll No. .....

B.C.A.-III Sem.

18012

B.C.A. Examination, Dec.-2020

DATA STRUCTURE USING C AND C++

(BCA-302)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section- A

(Very Short Answer Questions)

Note : Attempt **all** questions. 3×5=15

1. Explain the data structure's operations. 3

2. How can we minimize the stack overflow? 3

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3. Write prefix & postfix form for A+B *(C-D) / (E-F) 3

4. Design a recursive factorial function using C/C++ language. 3

5. Explain the term Hashing. 3

Section - B

(Short Answer Questions)

Note: Attempt any **two** questions. 2×7½=15

6. Describe the types of sparse matrix. How can we store a 2D sparse matrix in a corresponding single dimensional array? Find the formula for address calculation. 7½

7. Explain D-Queue & priority queue with a suitable example. 7½

8. Write a program in C/C++ to multiply two matrices A & B. 7½

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Section - C

(Detailed Answer Questions)

Note: Attempt any **three** questions. 3×15=45

9. (a) Perform Quick sort operation on given numbers. **15**

   66, 35, 48, 55, 62, 77, 25, 38, 18, 40, 30, 20.

   (b) Apply Bubble sort on DATASTRUCTURES.

10. Discuss the programming code in C/C++ language to create, insert & delete the elements in a singly linked list. **15**

11. Explain the properties of B-Trees. Also create a B-Tree of order 3 for following data. **15**

    Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec.

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12. (a) Create a Heap tree with the following element. 15

    95, 13, 12, 71, 96, 10, 62, 43, 35, 38.

    (b) Make a Binary search tree for given data.

    14, 15, 4, 9, 7, 18, 3, 5, 16, 4, 20, 17, 9, 14, 5.

13. (a) Design a function CQINSERT for static implementation of circular queue. 15

    (b) Differentiate linear & Binary search with suitable example.

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(21119)

B.C.A.-III Sem.

Printed Pages : 3

Roll No. .....

18012

B.C.A. Examination, November-2019

DATA STRUCTURE USING C AND C++

(BCA-302)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from ***all*** sections as per instructions.

Section–A

(Very Short Answer Questions)

Note : Attempt all the ***five*** questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 5×3=15

1. Write the limitations of arrays.

2. Define stacks and queues with an example.

3. What is the advantage of a header node in a linked list ?

4. How a binary tree is traversed in C language ?

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5. What is the basic concept of insertion sorting ?

Section–B

(Short Answer Questions)

Note: Answer any ***two*** questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words.

2×7½=15

6. What do you mean by sparse matrix ? Explain how a sparse matrix is represented in memory.

7. What is D-queue ? Explain the insertion and deletion operations with the help of suitable example.

8. Write an algorithm to delete last node from a linked list.

Section–C

(Detailed Answer Questions)

Note: Attempt any ***three*** questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 3×15=45

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9. Write algorithm and its C syntax to insert an element at the Kth position into the linear array.

10. Write an algorithm to evaluate postfix expression and also implement the algorithm to the following expression :

3, 1, +, 2, ↑, 7, 4, –, 2, * , +, 5, –

11. Write algorithm to perform insertion and deletion operations on binary trees and explain them with an example. https://www.ccsustudy.com

12. Describe hashing and various hashing techniques in detail.

13. Explain the following :

(i) Priority Queues

(ii) Heap Sort

(iii) Applications of Binary Search Tree

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(21218)

BCA- III Sem.

Roll No. .....

18012

B. C. A. Examination, Dec. 2018

Data Structure Using C and C++

(BCA-302)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Answer all the *five* questions. Each question carries 3 marks. Very short answer is required. 3×5=15

1. What are AVL trees ?

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2. What do you understand by Data Structure ?

3. What is an array ? Explain.

4. Give the application of stack.

5. Define circular list and chain.

Section-B

(Short Answer Questions)

Answer any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. Write the procedure to delete an element from an array.

7. Write a recursive function to count the number of leaves in a binary tree.

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8. Write a function to reverse a stack using push and pop operation.

Section-C

(Detailed Answer Questions)

Answer any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. (a) How to delete an element from a linked queue ? Write procedure.

   (b) Find the time complexity of Bubble Sort Algorithm.

10. (a) What is Binary search tree ?

    (b) What is a B-tree ?

11. (a) Write a function to evaluate a postfix expression. Show status of stack for each scanned token if the postfix expression is :

    ABC * DEF $/- G+H* –

    (b) Differentiate between stacks and queues.

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12. Write a program to print a given singly linked list in the reverse order.

13. Write short notes on the following :

    (a) Priority Queues

    (b) Hashing Techniques.

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(21218)

BCA- III Sem.

Roll No. .....

18012

B. C. A. Examination, Dec. 2018

Data Structure Using C and C++

(BCA-302)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Answer all the *five* questions. Each question carries 3 marks. Very short answer is required. 3×5=15

1. What are AVL trees ?

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2. What do you understand by Data Structure ?

3. What is an array ? Explain.

4. Give the application of stack.

5. Define circular list and chain.

Section-B

(Short Answer Questions)

Answer any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. Write the procedure to delete an element from an array.

7. Write a recursive function to count the number of leaves in a binary tree.

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8. Write a function to reverse a stack using push and pop operation.

Section-C

(Detailed Answer Questions)

Answer any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. (a) How to delete an element from a linked queue ? Write procedure.

   (b) Find the time complexity of Bubble Sort Algorithm.

10. (a) What is Binary search tree ?

    (b) What is a B-tree ?

11. (a) Write a function to evaluate a postfix expression. Show status of stack for each scanned token if the postfix expression is :

    ABC * DEF $/- G+H* –

    (b) Differentiate between stacks and queues.

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12. Write a program to print a given singly linked list in the reverse order.

13. Write short notes on the following :

    (a) Priority Queues

    (b) Hashing Techniques.

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N (Printed Pages 4)

(201217) Roll No. .....

B.C.A. - III Sem.

18012

B.C.A. Examination, Dec.- 2017

Data Structure Using C and C++

(BCA-302)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all sections as per Instructions.

Section-A

(Very Short Answer Questions)

Note : Attempt all the **five** questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. $3 \times 5 = 15$

1. How a two-dimensional array is represented in memory?

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2. Discuss the significance of priority queues.
3. How the end-of-list condition will be tested in a circular linked list?
4. Differentiate between preorder and post order tree traversal.
5. What is the concept of merge sorting?

Section-B

(Short Answer Questions)

Note : Answer any **two** questions out of the following **three** questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. $7\frac{1}{2} \times 2 = 15$

6. Write a program in C to insert an item of information as the first node in the linked list.
7. What is B-tree? How do you construct the B-tree? Explain with example.
8. What do you mean by linear search? Discuss the complexity of linear search.

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Section-C

(Detailed Answer Question

Note : Answer any **three** questions [illegible] of the following **five** questions. Ea[illegible] question carries 15 marks. Answer is [illegible]uired in detail. [illegible]5×3=45

9. Discuss the following with exam[illegible]:
   (a) Lower triangular matrix
   (b) Upper triangular matrix
   (c) Tridiagonal matrix

10. How a stack is represented in [illegible] array? Describe the various application [illegible] stacks. Explain prefix, infix and postfix ex[illegible]ssions with the help of examples.

11. Write an algorithm to delete [illegible]e K^{th} node from a two-way linked list. E[illegible]in the algorithm by taking an example.

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12. What is binary search tree? The following list of letters are inserted into an empty binary search tree:

    J R D G T E M H P A F Q

    (a) Find the final tree T,
    (b) Find the Post-order traversal of T.

13. Write an algorithm for heap sort and implement the algorithm to sort the following numbers:

    42, 32, 52, 22, 77, 66, 88

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N

(21216)

B. C. A.- III Sem.

Roll No. .....

18012

B. C. A. Examination, Dec. 2016

Data Structure Using C and C++

(BCA-302)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Answer all the *five* questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 3×5=15

1. How single dimensional array can be represented in computer memory ?

2. What is priority queue ?

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3. Give the use of header node in linked list.

4. Differentiate between linear search and binary search technique.

5. Briefly explain complete binary tree with example.

Section-B

(Short Answer Questions)

Answer any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. 7½×2=15

6. What is postfix expression form of any infix expression ? Write an algorithm to convert infix expression into postfix expression.

7. Write a program in C language to search a given element in linked list.

8. Explain the various types of hash functions with example.

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Section-C

(Detailed Answer Quest

Answer any *three* questions out following five questions. Each question carries arks. Answer is required in detail. 15×3=45

9. What do you mean by traversal ary tree ? Give the recursive algorithm for var ypes of binary tree traversal with suitable exam

10. Explain the following terms in re with queue :
    (a) Insertion and deletion oper
    (b) Applications
    (c) Limitations and their reme

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11. What is binary search tree ? Writ gram in C++ to create a binary search tree havin lements. Apply the given program on the followi a :

    –5, 10, 100, 5, –6, 1, 0

12. Explain Merge Sort with a suitab mple in detail.

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13. Write notes on the following (any three) :
    (a) Two-way list
    (b) Selection sort
    (c) Sparse array with example
    (d) Linear search.

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(20116) Roll No. .....

B.C.A. - III Sem.

18012

B.C.A. Examination, Dec. 2015

Data Structure using C and C++

(BCA-302)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section - A

(Very Short Answer Questions)

Note : Attempt all **five** questions. Each question carries 3 marks. Answers should not exceed 75 words.

1. Write a short note on sparse Array.
2. Enlist various primitive operations of stack.

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3. How and why headers are important in Linked List?
4. What do you understand by Indexing in Binary Search Trees?
5. Describe how hashing enhances the system performance.

Section - B

(Short Answer Questions)

Note : 1. Attempt any **two** questions from 3 Questions.

2. Each Question carries 7.5 marks each.
3. Answers should not exceed 200 words.

6. Briefly describe the concept of Tridiagonal matrices with examples.
7. Differentiate between D-queues and priority queues.
8. Write an algorithm to perform Merge Sort.

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Section - C

Note : 1. Attempt any **three** questions from **five** questions.

2. Each question carries 15 marks.

3. Answers should be in detail. $3 \times 15 = 45$

9. 'P' is a two dimensional Array with 5 rows and 5 columns i.e. P[1:5, 1:5]. Each element of array is stored in three memory locations. If P[1, 1] begins at address 800. Find the location of P[4, 3] when :

   (i) Organization is row major.

   (ii) Organization is column major.

10. Define Stack. Write algorithms for PUSH and POP operations. Also implement stack by using arrays.

11. Define overflow and underflow conditions of Linked list. Write an algorithm to insert a node into linked list at a specific location.

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12. Define the following: $3 \times 5 = 15$

   (i) Inverted Trees

   (ii) Post order traversal

   (iii) Binary Search Tree (BST)

13. Create a heap for the list of elements given below :

Index	Values
1	97
2	22
3	43
4	100
5	34
6	97
7	81
8	10
9	44
10	41

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M (Printed Pages 4)

(20115) Roll No. .....

BCA - III Sem.

18012

BCA Examination, Dec. 2014

Data Structure using 'C' and C^{++}

(BCA-302)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note: Attempt **all** the Sections as per instructions.

Section-A

(Very Short answer Questions)

Note: Attempt all the **five** questions. Each question carries **03** marks. Very short answer is required not exceeding 75 words.

1. How 2-dimensional arrays can be represented? 03

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2. What is D-queue? 03

3. Give the linked representation of any list? 03

4. Differentiate between searching and sorting? 03

5. Briefly explain Full binary tree with example. 03

Section-B

(Short answer questions)

Note: Attempt any **two** questions out of the following 03 questions. Each question carries 7.5 marks. Short answer is required not exceeding 200 words.

6. What are various operations that can be performed on linked list? Write an algorithm to insert a given element at the beginning position of given linked list. 7.5

7. What is B-Tree? How B-Tree can be created? Explain with a suitable example. 7.5

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8. What is prefix form of any expression? Write a program in C Language to convert any given infix expression into prefix form. 7.5

Section-C

(Detailed answer questions)

Note: Attempt any **three** questions out of the following **5** questions. Each question carries **15** marks. Answer is required in detail.

9. Explain the following Binary Tree Traversal algorithm with suitable example: 15

   (a) Inorder

   (b) Pre order

   (c) Post order

10. Explain following terms in relation with stack:

    (a) Push operation with example 6

    (b) Pop operation with example 6

    (c) Applications 3

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11. What do you meant by searching? Explain Binary Search technique with a suitable example. 15

12. Explain Heap Sort with a suitable example. 15

13. Write a note on following (any **three**): $3 \times 5 = 15$

    (a) Evaluation of Postfix expression

    (b) Linear Search

    (c) Header Linked List

    (d) Vector representation of Sparse arrays

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(21213) Roll No. .....

BCA-III Sem.

18012

B. C. A. Examination, Dec. 2013

Data Structure Using C & C++

(BCA-302)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt all the Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 3×5=15

1. What do you mean by transpose of an array ?

2. How priority queues are different from simple queue ?

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3. Diagrammatically represents various types of linked list.

4. What is indexing in Binary Search Tree (BST) ?

5. What is row major array ? List its applications.

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. 7½×2=15

6. What do you mean by trace of a matrix ? Design an algorithm.

7. Write a program that takes as input an integer number, split the number in its digits and stores the digits in a linked list structure.

8. Write short note on linear probing.

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Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. Write a function to delete a node from a circular linked list.

10. Transform the following infix expression into postfix form :

$$Z + (y * x - (w/v \wedge u) * t) * S.$$

11. Differentiate between (with examples) :

(i) Binary tree and balanced binary tree

(ii) Column major and row major arrays

(iii) Single way list and two way list.

12. Draw the binary tree that represents the following infix expression :

$$(((x-y)+z)/((u-v)*w))+t$$

and for the following prefix expression :

$$+ * x y | - z \vee w.$$

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13. How insertion sort is different from selection sort ? Explain with help on algorithms and examples. Also enumerate the best sort.

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